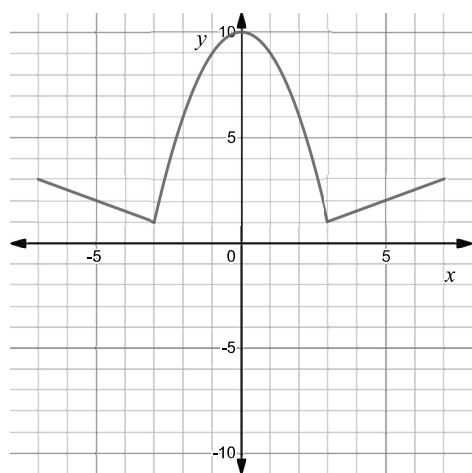


Precalc Lesson 1.6 Homework

Answer Key

1. Neither, because g is not symmetric about the y -axis or the origin.
2. Any three of the following: $(5, -10)$, $(-3, 4)$, $(-11, -8)$, $(9, 1)$, $(-4, 0)$
3. g is even because the parent function $y = x^2$ is even and shifting vertically and stretching does not change the symmetry of the graph.
OR g is even because $g(-x) = 3(-x)^2 - 7 = 3x^2 - 7 = g(x)$.
4. f is neither even nor odd because $f(-x) = (-x)^3 + 5(-x)^2 + 11 = -x^3 + 5x^2 + 11$, which is not equal to $f(x)$ or $f(-x)$.
5. f is odd because $f(-x) = (-x)^3 + 5(-x) = -x^3 - 5x = -f(x)$.

6.



7. False. $y = (x - 3)^2$ is quadratic, but does not have y -axis symmetry.
8. One example is not enough to prove that a function is even. $f(x)$ is not equal to $f(-x)$ for all x . For example, $f(1) = 12$, but $f(-1) = 18$.

9. True. If a function is even then $f(x) = f(-x)$. Reflecting the function over the x -axis gives $-f(x) = -f(-x)$ which is still even because outputs at opposite x -values are still equal. If a function is odd then $f(x) = -f(-x)$. Reflecting over the x -axis gives $-f(x) = -(-f(-x)) = f(-x)$ which is still odd, because outputs at opposite x -values are opposite.
10. If a function is both even and odd then if $f(a) = b$, then we need that $f(-a) = b$ and $f(-a) = -b$. In order to be a function, $f(-a)$ can only have one output! This only works when $b = -b$ which only happens when $b = 0$. The line $y = 0$ is a function that is both even and odd.